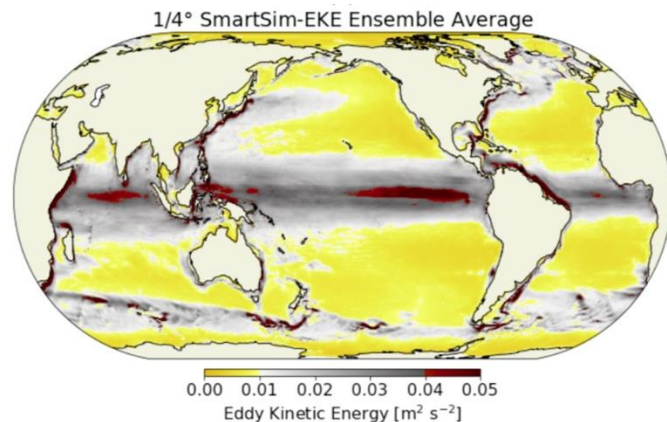


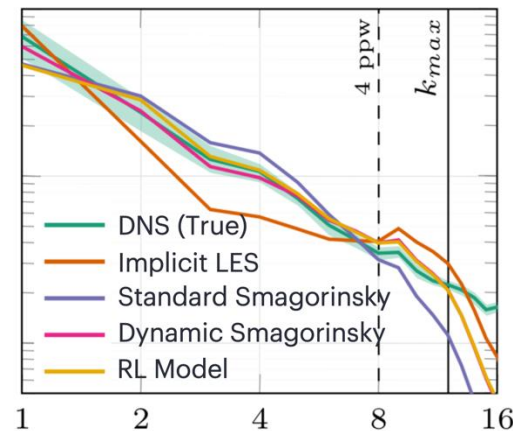
AI/HPC research HPE Labs

Andrew.Shao@hpe.com

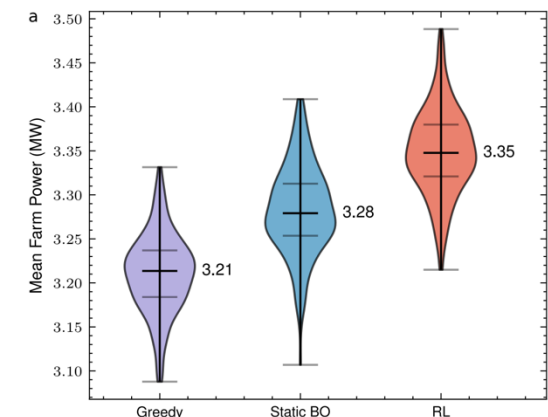
- HPE Labs performs fundamental research with external partners across the computer and domain sciences
 - PhD internships, postdoc, co-PIs
- Three research areas under my portfolio
 - What numerical techniques can be applied to AI surrogates of dynamic systems?
 - Where are the benefits of blending AI methods and simulation to open new areas of research?
 - How do these hybrid simulation/AI applications challenge our view of computing and what software needs to be developed?
- Numerical techniques and AI (UVic, RWE)
 - Multi-timescale aware loss functions for training surrogate models of coupled earth system
- Blending AI/Simulation (DOE, LRZ)
 - Active learning and Bayesian methods for experimental design
- Open-source software for AI/Simulation
 - Scalable Hybrid HPC/AI workflow tools for AI-in-the-loop/AI-around-the-loop motifs (Rutgers/DOE)
 - Strategies for AI workloads in Fortran (Berkeley, Nvidia, Cambridge)



Partee et al.[JCS, 2023]



Kurz et al. [Physics of Fluids, 2024]



Mole et al. [Arxiv, 2025]